ALPEREN KARACAN

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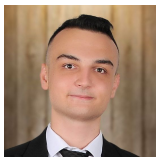

Summary

- AI Engineer specialized in computer vision, natural language processing, and deep learning solutions. Experienced in training, optimizing, and deploying scalable models using PyTorch. Designed models with high accuracy (%93) on custom datasets using advanced architectures such as EfficientNetV2-L and YOLOv8. Skilled in integrating AI solutions into real-world applications with strong experience in data analysis, model building, and full-stack implementation.

Education

Kocaeli University *Sep 2021 – Jun 2025*

Bachelor of Science in Computer Engineering

Work Experience

Huawei

Data Engineer Intern Jul 2025 - Aug 2025

- Assisted in data collection, processing, and management workflows.
- Contributed to improving processes by collaborating with the team.
- **Technologies:** Python, SQL, Linux

Kocaeli University

AI Project Intern Jul 2024 - Aug 2024

- Developed an object detection application using YOLOv8.
- Trained a model on the BDD100K dataset to detect traffic lights, signs, pedestrians, and vehicles.
- Implemented both image- and video-based detection with visualization of model outputs.
- **Technologies:** Python (PyTorch, OpenCV, NumPy, PIL), YOLOv8

Skills

Programming Languages: Python, Dart, C, Java

Projects

Hybrid AI for Nutritional Meal Analysis (Mobibes)

Python(Flask), PyTorch, YOLOv8, Flutter, MongoDB

- Built a mobile app for AI-powered nutritional tracking and meal analysis.
- Designed and trained an EfficientNetV2-L model (15K images / 40 classes) and used the Gemini API for food recognition and portion estimation.
- Used YOLOv8 to detect and segment multiple plates in an image.
- Developed Mobibes Assistant and a personalized meal planner powered by the Gemini API.
- Implemented email verification, password reset, and multi-language support.
- Built an admin panel for user monitoring and usage analytics.

Smart Quiz Platform

Python(FastAPI), Flutter, Gemini API

- Built a Gemini API-based service that generates different question types from user-provided text/documents (PDF, DOCX, TXT).
- Implemented student/teacher/admin roles and email verification.
- Developed classroom management and homework assignment features for teachers.
- Designed an exam mode, goal setting, and a detailed analytics module for students.
- Integrated gamification features including points, badges, and a leaderboard.
- Enabled exporting generated exams to PDF and Word formats.

AI-Powered CV & Career Platform

Python(FastAPI), Gemini API, MongoDB, Speech-to-Text

- Developed a platform that lets users create CVs from scratch, parse existing CVs using AI to auto-fill relevant fields, and edit information manually.

- Built an assistant that generates tailored cover letters, identifies skill gaps for target roles, and recommends courses/projects to close those gaps.
- Built a Gemini API-based ATS analyzer that visualizes metrics such as ATS compatibility, keyword matching, and content quality.
- Developed an application tracker and an interview simulator.

112 Emergency Call Center AI Agent

Python (Pydantic AI, FastAPI), Whisper, Gemini API

- Built a real-time 112 call center simulation; the system analyzes incoming calls with a hybrid (rule-based + LLM) approach to determine urgency.
- Designed a multi-agent pipeline that covers operator, triage, and dispatch roles.
- Developed tools for critical tasks (dispatch coordination, first-aid instructions, location verification) and orchestrated them with an LLM.
- Integrated Whisper (STT) and TTS to process voice calls and generate spoken responses.
- Built a FastAPI + WebSocket dashboard to monitor decision processes and agent interactions in real time.

Personal Finance Management App

Flutter, Hive, Gemini API

- Built an offline-first mobile app featuring income/expense tracking, subscriptions, bills, budgeting, savings modules, smart notifications, and a Gemini AI financial advisor.

Fraud Detection

Python (FastAPI), PyTorch, XGBoost, Optuna, MongoDB

- Built a real-time financial fraud detection system using XGBoost and Graph Neural Networks (GNN).
- Integrated SHAP for explainable AI (XAI) and a WebSocket-based interactive simulation UI.

Medication & Water Tracking App

Python (FastAPI), Flutter, MongoDB, Gemini API

- Built a mobile app for tracking medication, vitamins, and water intake with reminder notifications.
- Integrated an AI module that extracts medication information automatically via pill box image analysis.
- Designed a premium UI with statistics charts, pharmacy finder, and theme support.

Academic Article Search (Web Scraping)

Python, MongoDB, Elasticsearch

- Scraped academic article data from DergiPark.
- Implemented fast keyword search and filtering with Elasticsearch.

Academic Article Recommendation System (NLP)

Python, MongoDB, SciBERT, FastText

- Built a web application that recommends personalized articles based on user interests.
- Used FastText and SciBERT for content analysis.

Animal Classification Desktop App

Python, PyTorch, OpenCV, CustomTkinter, Scikit-learn

- Built a desktop app (CustomTkinter) to classify 90 animal species from images, folders, and live webcam input.
- Trained a Swin Transformer model using data augmentation, AMP, and early stopping.
- Added real-time predictions and one-click Google search for results.

User Interest Analysis

Python

- Analyzed posts from 50,000 users to extract interests from keywords and map them to categories.

Real-Time Order & Inventory Management

Python (Flask), MongoDB

- Built a web-based customer/product management and concurrent order processing system.
- Used multithreading for real-time stock updates.

References

- **Umut Can Apaydın** – Huawei, Engineering Manager
- **Zekeriya Sarı** – Huawei, Data Engineer Tech Lead
- **Uğur Özcan** – Huawei, Senior Data Engineer
- Contact information will be provided upon request.